

## CENG381 Stochastic Processes

### Random Walks on Graphs - Answer Key 1

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#### Q1 Solution

##### a) Transition probability:

From vertex  $x$ , the walk moves to each neighbour  $y$  of  $x$  with equal probability. There are  $\deg(x)$  neighbours, so:

$$P(x, y) = 1/\deg(x) \quad \text{if } y \text{ is a neighbour of } x$$

$$P(x, y) = 0 \quad \text{otherwise}$$

##### b) Stationary distribution $\pi(x) = \deg(x) / (2|E|)$ :

We verify  $\pi * P = \pi$ . For any vertex  $y$ :

$$\begin{aligned} (\pi * P)(y) &= \sum_{\{x\}} \pi(x) * P(x, y) \\ &= \sum_{\{x \sim y\}} [\deg(x)/(2|E|)] * [1/\deg(x)] \\ &= \sum_{\{x \sim y\}} 1/(2|E|) \\ &= \deg(y) / (2|E|) \\ &= \pi(y) \quad (\text{OK}) \end{aligned}$$

where the sum is over all  $x$  adjacent to  $y$  (there are  $\deg(y)$  such  $x$ ).

$$\text{Also } \sum_x \pi(x) = \sum_x \deg(x) / (2|E|) = 2|E|/(2|E|) = 1 \quad (\text{OK})$$

(using the handshaking lemma: sum of degrees =  $2|E|$ ).

##### c) Star graph: hub H, leaves A, B, C, D:

$$\text{Degrees: } \deg(H) = 4, \deg(A) = \deg(B) = \deg(C) = \deg(D) = 1$$

$$\text{Number of edges: } |E| = 4$$

Transition matrix:

From H: go to A, B, C, or D each with prob  $1/4$

From A: go to H with prob 1 (only neighbour)

(same for B, C, D)

|   | H   | A   | B   | C   | D     |
|---|-----|-----|-----|-----|-------|
| H | [ 0 | 1/4 | 1/4 | 1/4 | 1/4 ] |
| A | [ 1 | 0   | 0   | 0   | 0 ]   |
| B | [ 1 | 0   | 0   | 0   | 0 ]   |
| C | [ 1 | 0   | 0   | 0   | 0 ]   |
| D | [ 1 | 0   | 0   | 0   | 0 ]   |

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Stationary distribution:

$$\pi(H) = 4/(2 \cdot 4) = 4/8 = 1/2$$

$$\pi(A) = \pi(B) = \pi(C) = \pi(D) = 1/(2 \cdot 4) = 1/8$$

Interpretation: the hub is visited 50% of the time in the long run.

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### Q2 Solution

#### a) 5-cycle transition matrix (states 0,1,2,3,4):

|   | 0     | 1   | 2   | 3   | 4     |
|---|-------|-----|-----|-----|-------|
| 0 | [ 0   | 1/2 | 0   | 0   | 1/2 ] |
| 1 | [ 1/2 | 0   | 1/2 | 0   | 0 ]   |
| 2 | [ 0   | 1/2 | 0   | 1/2 | 0 ]   |
| 3 | [ 0   | 0   | 1/2 | 0   | 1/2 ] |
| 4 | [ 1/2 | 0   | 0   | 1/2 | 0 ]   |

#### b) Verify $\pi(j) = 1/5$ is stationary:

$$\begin{aligned}(\pi * P)(y) &= \sum_j \pi(j) * P(j, y) \\&= (1/5) * \sum_j P(j, y) \\&= (1/5) * (\text{column sum of } P \text{ at column } y)\end{aligned}$$

Each column of  $P$  sums to 1 (each state  $y$  has exactly 2 neighbours, each contributing  $1/2$ ):

$$\text{column sum} = 1/2 + 1/2 = 1$$

$$(\pi * P)(y) = (1/5) * 1 = 1/5 = \pi(y) \quad (\text{OK})$$

So  $\pi = \text{uniform}(1/5)$  is stationary for the 5-cycle.

#### c) Periods - odd vs even cycles:

5-cycle (odd  $n=5$ ):

From state 0, returns are possible in steps 2, 4, 5, 6, ...

In 2 steps:  $0 \rightarrow 1 \rightarrow 0$  or  $0 \rightarrow 4 \rightarrow 0$  (OK)

In 5 steps:  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0$  (OK)

$\gcd(2, 5) = 1 \Rightarrow \text{period } d = 1$  (aperiodic!)

4-cycle (even  $n=4$ ):

Returns from state 0 happen only in even steps: 2, 4, 6, ...

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(must travel an even number of edges to return on an even cycle)

$$\gcd(2, 4, \dots) = 2 \Rightarrow \text{period } d = 2$$

Key insight: odd cycles are aperiodic; even cycles have period 2.

Adding self-loops (lazy version) fixes even-cycle periodicity.

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### Q3 Solution

#### **a) Verify $f(x,z) = (x+z) \bmod 3$ for 3-cycle:**

$$\text{Lambda} = \{-1, +1\}, P(Z = -1) = P(Z = +1) = 1/2.$$

For any state  $x$  in  $\{0,1,2\}$  and any state  $y$  in  $\{0,1,2\}$ :

$$P\{f(x, Z) = y\} = P\{(x + Z) \bmod 3 = y\}$$

Case  $y = x+1 \bmod 3$ : need  $Z = +1$ , prob =  $1/2 = P(x, x+1 \bmod 3)$  (OK)

Case  $y = x-1 \bmod 3$ : need  $Z = -1$ , prob =  $1/2 = P(x, x-1 \bmod 3)$  (OK)

Case other  $y$ : impossible since  $Z$  in  $\{-1, +1\}$ , prob = 0 (OK)

This matches the 3-cycle transition matrix, so  $f$  is a valid random mapping representation.

#### **b) $(X_0, X_1, \dots)$ is a Markov chain:**

Given  $X_0, X_1, \dots, X_{\{n-1\}} = x_{\{n-1\}}$ , we have:

$$P(X_n = y \mid X_0, \dots, X_{\{n-1\}}) = P(f(X_{\{n-1\}}, Z_n) = y \mid X_{\{n-1\}} = x_{\{n-1\}})$$

$$= P(f(x_{\{n-1\}}, Z_n) = y) \quad [\text{since } Z_n \text{ is independent of past}]$$

$$= P(x_{\{n-1\}}, y) \quad [\text{by the representation property}]$$

This depends only on  $X_{\{n-1\}}$ , not the full history  $\Rightarrow$  Markov property.

The transition matrix is exactly  $P$ , and initial dist is  $\mu$ . (OK)

#### **c) General $n$ -cycle representation:**

$$f(x, z) = (x + z) \bmod n, \quad Z \text{ uniform on } \{-1, +1\}.$$

For any  $j$  in  $Z_n$ :

$$P(j, j+1 \bmod n) = P(Z=+1) = 1/2$$

$$P(j, j-1 \bmod n) = P(Z=-1) = 1/2$$

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$P(j, k) = 0$  for all other  $k$

This is exactly the random walk on the  $n$ -cycle transition matrix.

The key insight is that the i.i.d. sequence  $(Z_n)$  generates the chain via the deterministic function  $f$ , which is often more convenient for analysis and simulation.